

MCGUIRE AFB, NJ, USA

WMO: 724096

Lat: 40.017N

Lon: 74.600W

Elev: 40

StdP: 100.85

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14706

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.0	-9.4	-19.8	0.6	-10.0	-17.3	0.8	-7.7	12.3	1.7	11.3	0.4	4.2	280	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	33.8	24.3	32.4	23.7	31.0	23.0	26.0	30.8	25.2	30.0	24.4	28.8	4.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	19.9	28.1	23.8	18.7	27.3	22.9	17.7	26.5	80.5	30.9	76.7	30.0	73.5	28.7	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	9.0	8.0	-15.1	36.4	2.6	2.0	-16.9	37.8	-18.5	39.0	-19.9	40.1	-21.8	41.6
			-15.7	27.5	2.5	1.2	-17.5	28.4	-19.0	29.1	-20.4	29.8	-22.2	30.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	0.1	1.6	5.6	11.6	16.8	21.8	24.7	19.9	13.8	7.8	3.2	(6)	
(7)		DBStd	9.60	5.89	5.31	5.07	4.79	4.27	3.65	2.85	2.87	3.61	4.46	4.61	5.07	(7)
(8)		HDD10.0	1070	313	239	155	33	1	0	0	0	12	96	221		(8)
(9)		HDD18.3	2700	567	469	396	210	82	11	0	1	24	155	316	470	(9)
(10)		CDD10.0	2023	4	4	18	82	213	354	457	423	298	129	30	9	(10)
(11)		CDD18.3	611	0	0	1	8	36	116	198	166	72	14	1	0	(11)
(12)		CDH23.3	5448	0	1	10	104	368	1042	1872	1414	538	97	4	0	(12)
(13)		CDH26.7	2028	0	0	2	32	117	379	793	523	159	21	1	0	(13)
(14)	Wind	WSAvg	3.6	4.2	4.3	4.5	4.2	3.5	3.2	3.0	2.8	3.0	3.3	3.6	3.9	(14)
(15)	Precipitation	PrecAvg	1145	89	63	102	89	88	104	113	122	102	93	76	102	(15)
(16)		PrecMax	1461	177	116	217	255	172	218	253	372	211	275	166	204	(16)
(17)		PrecMin	806	43	15	31	32	24	34	29	28	21	17	20	27	(17)
(18)		PrecStd	155	32	26	45	46	34	53	42	69	54	54	36	40	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.7	20.1	24.2	30.5	32.4	34.6	36.1	35.3	32.9	29.1	22.7	19.9	(19)
(20)		MCWB	14.9	14.4	16.1	19.1	21.8	24.6	24.6	24.7	23.6	20.8	16.9	16.8		(20)
(21)		2%	DB	15.5	16.2	20.1	26.3	30.0	32.5	34.1	32.8	30.6	26.1	20.0	16.6	(21)
(22)		MCWB	13.6	11.9	13.4	17.1	21.2	23.7	24.5	24.2	22.3	19.7	15.7	14.4		(22)
(23)		5%	DB	12.2	12.8	17.0	22.9	27.5	30.8	32.6	31.2	28.4	23.4	17.8	14.0	(23)
(24)		MCWB	9.9	9.3	11.9	15.6	20.0	23.0	24.0	23.4	21.7	18.4	14.6	12.0		(24)
(25)		10%	DB	8.8	9.9	14.1	20.1	25.0	28.9	31.0	29.8	26.5	21.3	15.7	11.4	(25)
(26)		MCWB	6.5	6.6	10.1	14.1	18.5	21.7	23.4	22.7	20.5	17.3	12.6	9.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.4	16.5	17.4	20.6	23.6	26.2	27.5	27.1	25.4	23.0	19.2	17.8	(27)
(28)		MADB	17.7	19.1	22.0	27.3	29.6	31.6	31.9	31.4	30.1	25.9	20.7	18.4		(28)
(29)		2%	WB	13.8	13.0	15.1	18.9	22.4	25.0	26.2	25.8	24.2	21.3	17.4	15.0	(29)
(30)		MADB	15.6	14.9	19.0	24.0	27.9	30.5	31.2	30.2	27.5	24.3	18.9	16.2		(30)
(31)		5%	WB	10.2	9.7	12.8	17.2	21.1	24.0	25.3	24.9	23.1	19.9	15.3	12.5	(31)
(32)		MADB	11.6	12.0	16.0	21.4	25.7	28.9	30.4	29.1	26.2	22.4	17.1	14.1		(32)
(33)		10%	WB	6.9	6.7	10.3	15.3	19.8	22.9	24.5	24.0	22.2	18.1	13.2	9.7	(33)
(34)		MADB	8.3	9.3	13.8	19.2	23.4	27.2	29.2	27.7	25.2	20.5	15.3	11.0		(34)
(35)	Mean Daily Temperature Range	MDBR	8.8	9.7	10.5	11.7	11.3	11.0	10.8	10.6	10.9	11.0	10.4	8.9		(35)
(36)		5% DB	MADB	12.4	14.4	15.3	15.7	14.5	12.9	12.5	12.0	12.8	13.2	13.0	11.6	(36)
(37)		MCWBR	10.5	10.7	10.0	8.2	7.1	5.8	4.7	5.0	6.0	7.4	9.1	10.4		(37)
(38)		5% WB	MADB	11.7	13.1	13.7	13.4	12.8	11.8	11.0	10.4	10.3	11.0	10.4	11.1	(38)
(39)		MCWBR	10.8	11.2	10.3	8.3	6.9	6.0	5.1	4.9	5.7	7.4	8.9	10.7		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.312	0.320	0.357	0.386	0.432	0.493	0.493	0.486	0.404	0.360	0.329	0.310	(40)	
(41)		taud	2.473	2.446	2.397	2.340	2.243	2.111	2.125	2.135	2.337	2.468	2.512	2.527	(41)	
(42)		Ebn at Noon	855	899	897	888	851	796	792	786	836	842	828	831	(42)	
(43)		Edh at Noon	78	92	108	122	137	157	153	148	113	90	75	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.94	2.85	3.79	4.88	5.40	5.93	6.05	5.23	4.35	3.09	2.20	1.69	(44)	
(45)		RadStd	0.17	0.30	0.31	0.42	0.62	0.51	0.29	0.34	0.44	0.31	0.20	0.17	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (66 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+119	+68	(47)

Nomenclature: See separate page